

Task 13 – Duplicate Record Check

Data Analytics Internship | Retail Sales Data

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1. What I did

I worked on a small retail-sales dataset to check whether any records had been entered more than once. The main aim was to identify genuine duplicate rows, understand how they could affect the analysis, and then create a cleaner version of the data. I also kept an audit note so that the cleaning process is easy to understand.

2. How I checked the data

I used two checks. First, I compared the complete rows to find records where every value was the same. Second, I checked the **Order_ID** column because it acts as an important identifier for each sale. I did not remove a record just because an ID appeared more than once; I first considered whether the other details were also identical.

3. What I found

Check	Result
Total records checked	24
Exact duplicate rows found	4
Duplicate groups	2
Duplicate Order IDs	RS1004 and RS1020
Records after cleaning	22

The check showed two exact duplicate groups. Order ID **RS1004** appears twice with exactly the same information, and the same happens with **RS1020**. Since these rows contain identical values, keeping both would count the same sale twice.

4. Duplicate records identified

Row	Order ID	Date	Customer	Product	Region	Sales	Qty
5	RS1004	2026-09-02	Delta Retail	USB Hub	South	999	1
6	RS1004	2026-09-02	Delta Retail	USB Hub	South	999	1
22	RS1020	2026-09-10	Urban Store	HDMI Cable	South	499	2
23	RS1020	2026-09-10	Urban Store	HDMI Cable	South	499	2

These four rows belong to two duplicate pairs. For the cleaned copy, I kept the first occurrence of each pair and removed the repeated copy.

5. Cleaning decision and audit note

I used a simple and safe rule: **only exact duplicate rows were removed automatically**. If the same Order_ID had different customer, product, date, or sales information, I would keep those records for manual business review instead of deleting them blindly. This is important because duplicate IDs do not always mean duplicate transactions.

Audit note: 24 records were checked. Four rows were part of exact duplicate pairs. After removing the repeated copies, the cleaned dataset contains 22 records. No record was deleted only because its Order_ID was repeated.

6. Cleaned data

Order ID	Date	Customer	Product	Category	Region	Sales	Qty
RS1001	2026-09-01	Aarav Traders	Laptop Bag	Accessories	North	1299	1
RS1002	2026-09-01	Bright Mart	Wireless Mouse	Accessories	North	799	2
RS1003	2026-09-02	City Store	Keyboard	Accessories	West	1499	1
RS1004	2026-09-02	Delta Retail	USB Hub	Accessories	South	999	1
RS1005	2026-09-03	Elite Shop	Monitor Stand	Office	East	1899	1
RS1006	2026-09-03	Fresh Point	Desk Lamp	Office	North	699	2
RS1007	2026-09-04	Galaxy Mart	Notebook	Stationery	West	299	3
RS1008	2026-09-04	Horizon Store	Pen Set	Stationery	South	199	5
RS1009	2026-09-05	Icon Retail	Keyboard	Accessories	East	1499	1
RS1010	2026-09-05	Jupiter Shop	USB Cable	Accessories	North	399	3
RS1011	2026-09-06	Kite Mart	Webcam	Electronics	West	2299	1
RS1012	2026-09-06	Lotus Store	Headphones	Electronics	South	1799	1
RS1013	2026-09-07	Metro Retail	Power Bank	Electronics	East	1299	1
RS1014	2026-09-07	Nova Shop	Laptop Sleeve	Accessories	North	899	1
RS1015	2026-09-08	Orbit Mart	Mouse Pad	Accessories	West	499	2
RS1016	2026-09-08	Prime Store	Desk Organizer	Office	South	599	2
RS1017	2026-09-09	Quick Buy	Sticky Notes	Stationery	East	149	4
RS1018	2026-09-09	Royal Retail	Calculator	Stationery	North	599	1
RS1019	2026-09-10	Smart Mart	Webcam	Electronics	West	2199	1
RS1020	2026-09-10	Urban Store	HDMI Cable	Accessories	South	499	2
RS1021	2026-09-11	Value Shop	Keyboard	Accessories	East	1499	1
RS1022	2026-09-11	Wave Retail	Keyboard	Accessories	East	1599	1

7. Conclusion

After checking the dataset, I found that duplicate entries were present and could have caused the sales figures to be counted more than once. By removing only the confirmed exact duplicates, I reduced the dataset from 24 to 22 records while keeping the cleaning process transparent. This gives a more reliable dataset for further analysis and dashboard creation.

8. Interview questions

Q1. What is a duplicate row?

A duplicate row is a record that repeats the same information as another record across the columns being checked.

Q2. How can duplicates affect totals?

Duplicates can make totals incorrect because the same transaction may be counted more than once. For example, total sales, quantity, and the number of transactions can become higher than their actual values.

Tools/Concepts: Excel duplicate checks, Pandas `duplicated()`, `drop_duplicates()`, key-column checking, data cleaning, and audit documentation.